A crystalline solid is one in which its constituents (atoms, ions or molecules), have an orderly arrangement extending over a long range. The arrangement of such constituents in a crystalline solid is such that the potential energy of the system is at minimum. In contrast, in amorphous solids (In Greek, amorphous means no form) the constituents are randomly arranged.

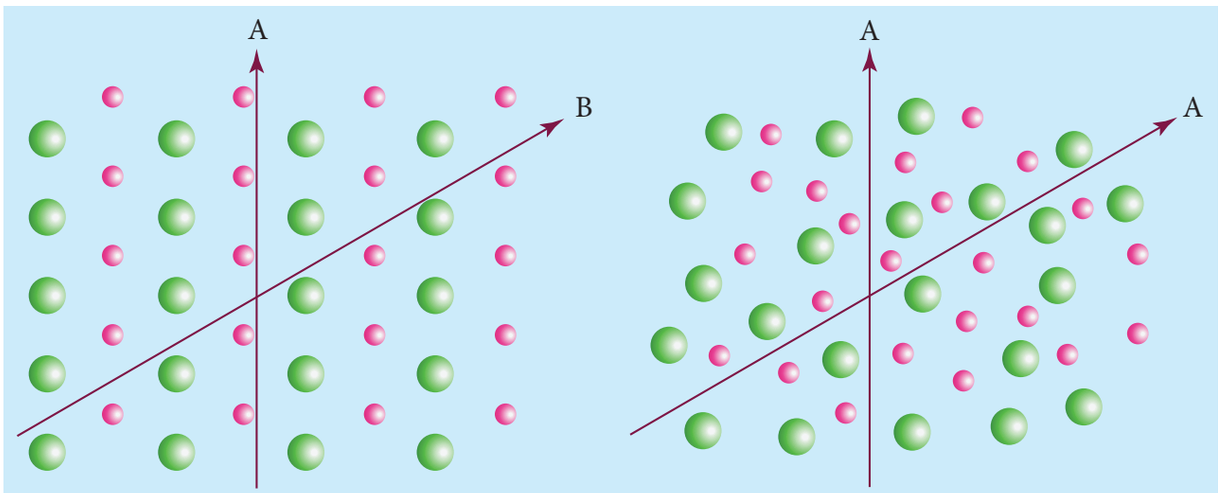
The following table shows the differences between crystalline and amorphous solids.

S.no	Crystalline solids	Amorphous solids
1	Long range orderly arrangement of constituents.	Short range, random arrangement of constituents.
2	Definite shape	Irregular shape
3	Generally crystalline solids are anisotropic in nature	They are isotropic* like liquids
4	They are true solids	They are considered as pseudo solids (or) super cooled liquids
5	Definite Heat of fusion	Heat of fusion is not definite
6	They have sharp melting points.	Gradually soften over a range of temperature and so can be moulded.
7	Examples: NaCl , diamond etc.,	Examples: Rubber , plastics, glass etc

Table 6.1 differences between crystalline and amorphous solids

*Isotropy

Isotropy means uniformity in all directions. In solid state isotropy means having identical values of physical properties such as refractive index, electrical conductance etc., in all directions, whereas anisotropy is the property which depends on the direction of measurement. Crystalline solids are anisotropic and they show different values of physical properties when measured along different directions. The following figure illustrates the anisotropy in crystals due to different arrangement of their constituents along different directions.



Anisotropy in Crystals **Isotropy in Amorphous solids**

XII U6 Solid State - Jerald.indd 178 3/29/2019 6:30:15 PM